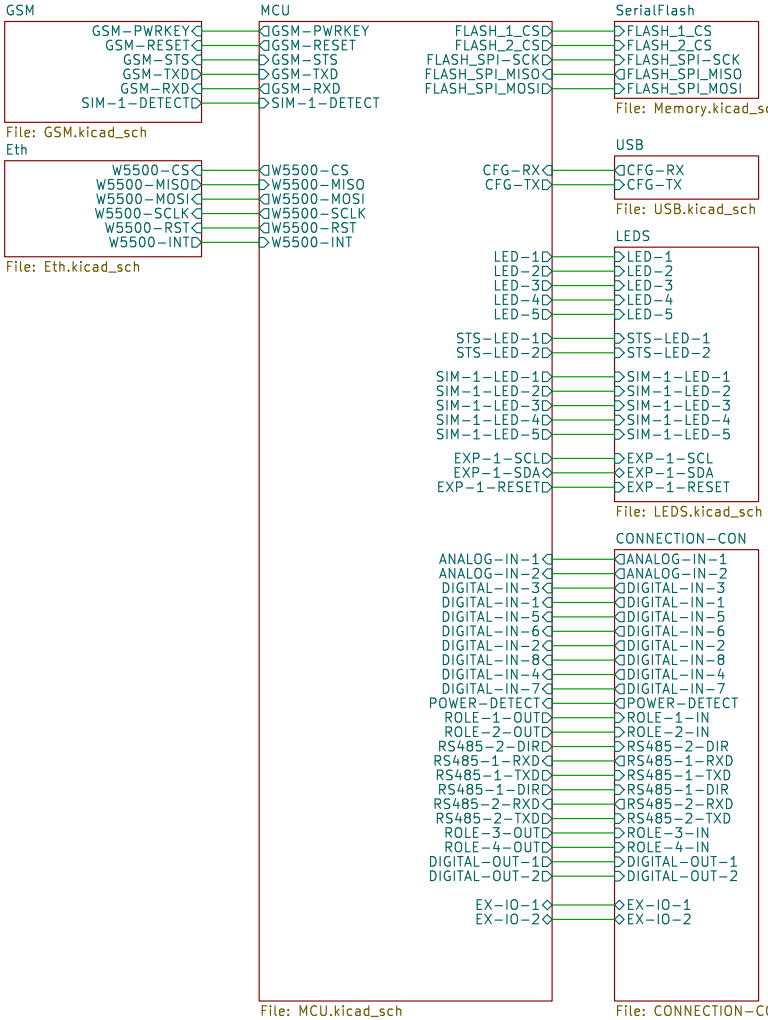


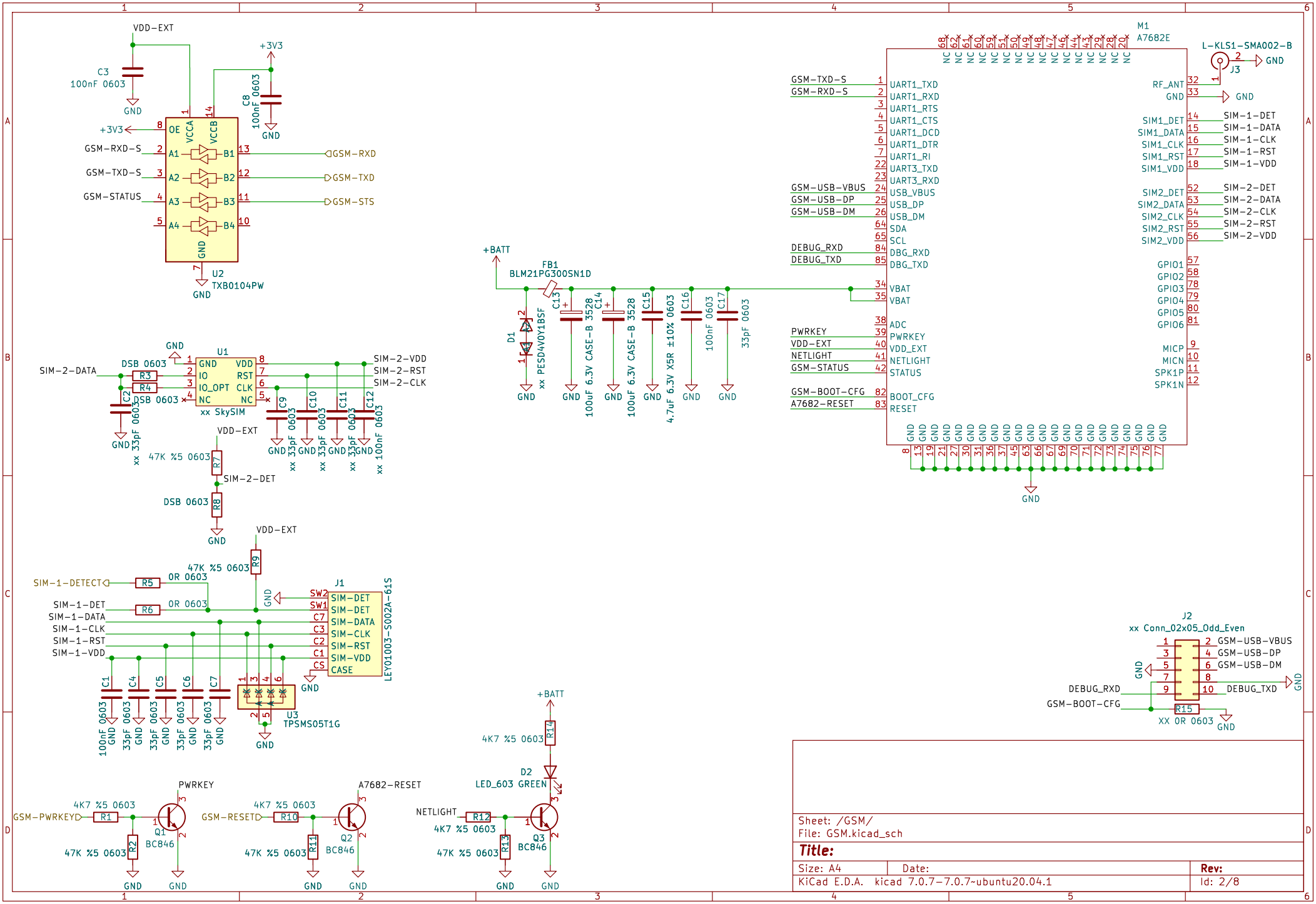
FID1
Fiducial
FID2
Fiducial
FID3
Fiducial

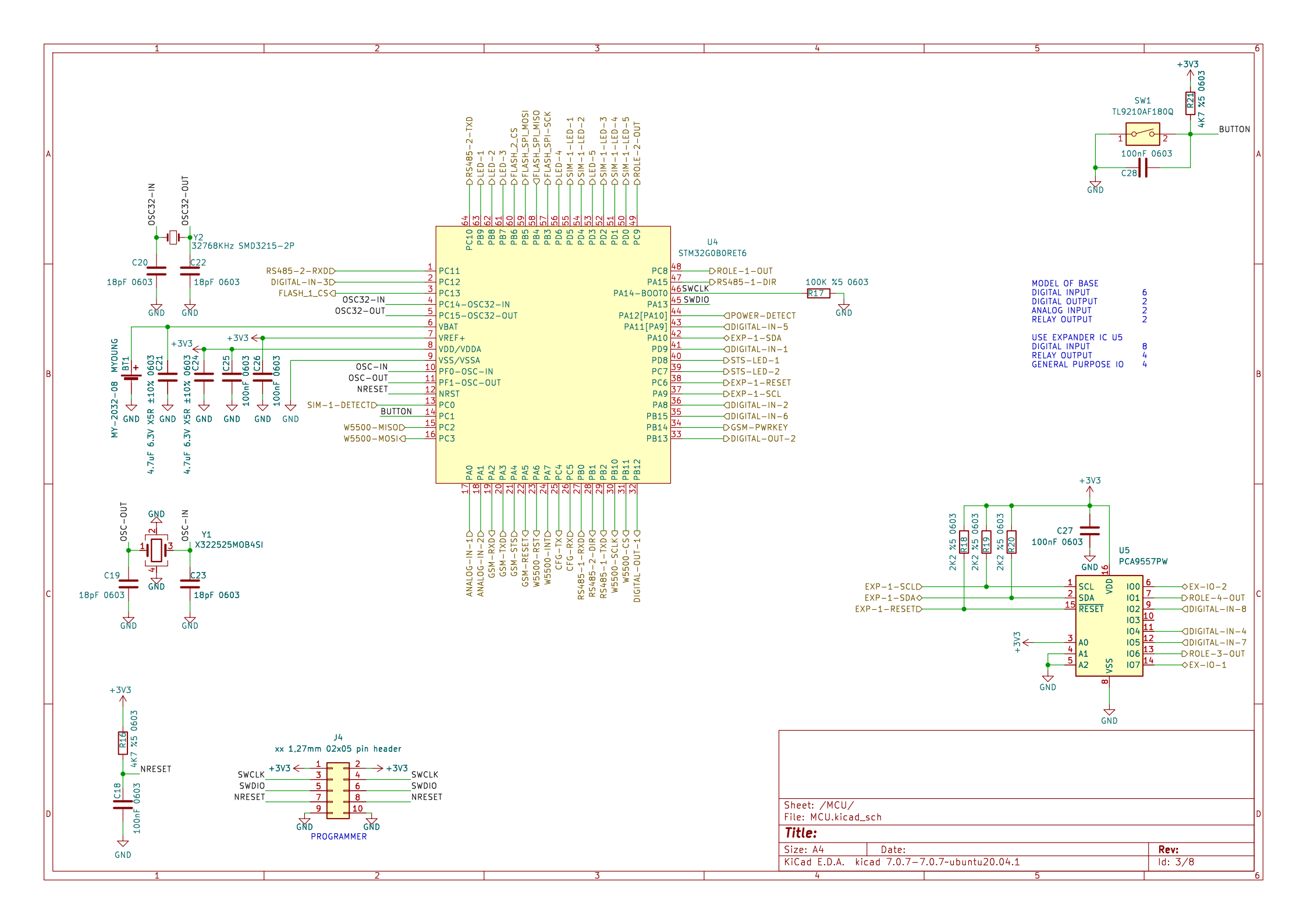
B1
BLU4GM_Board

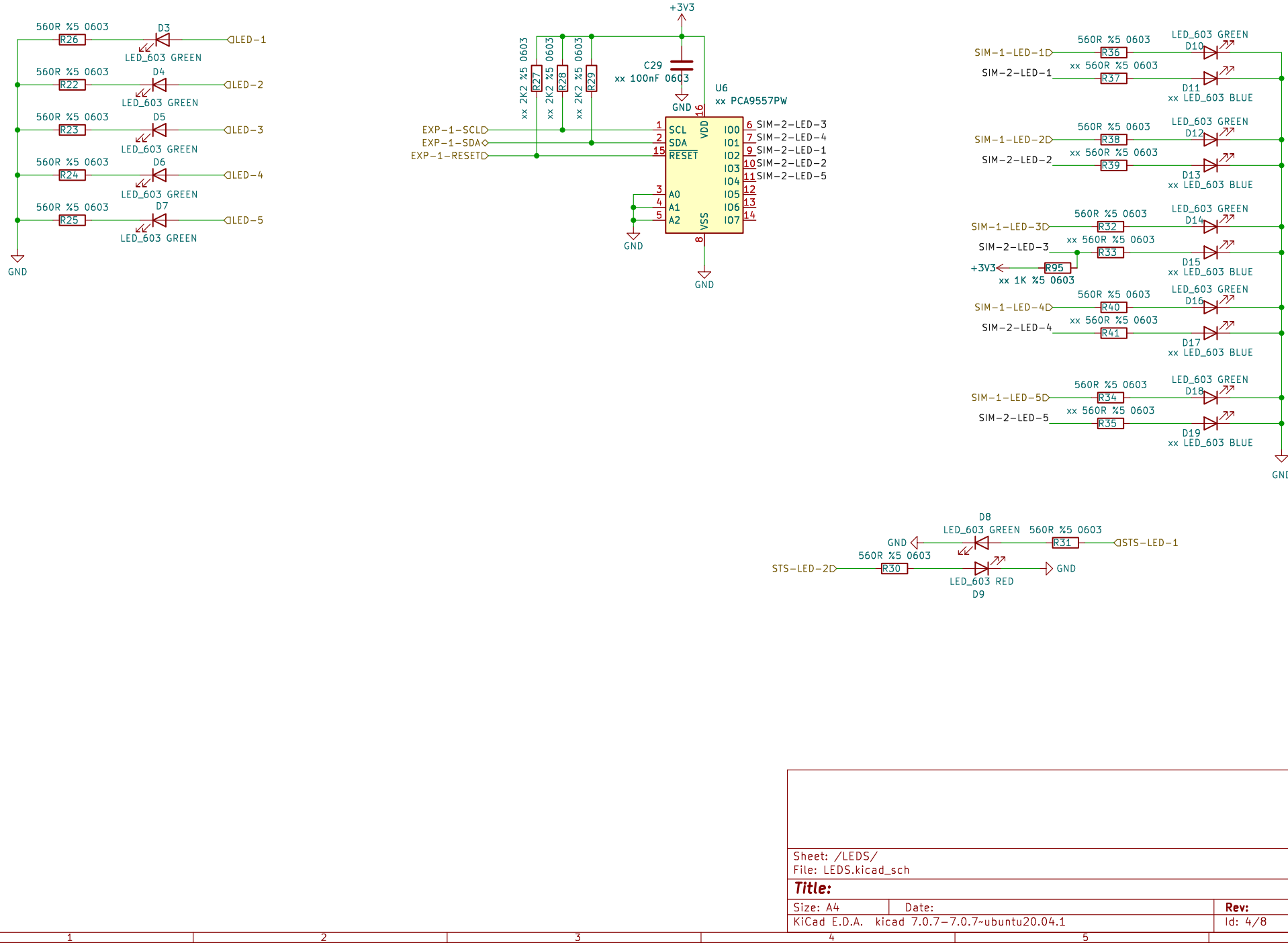
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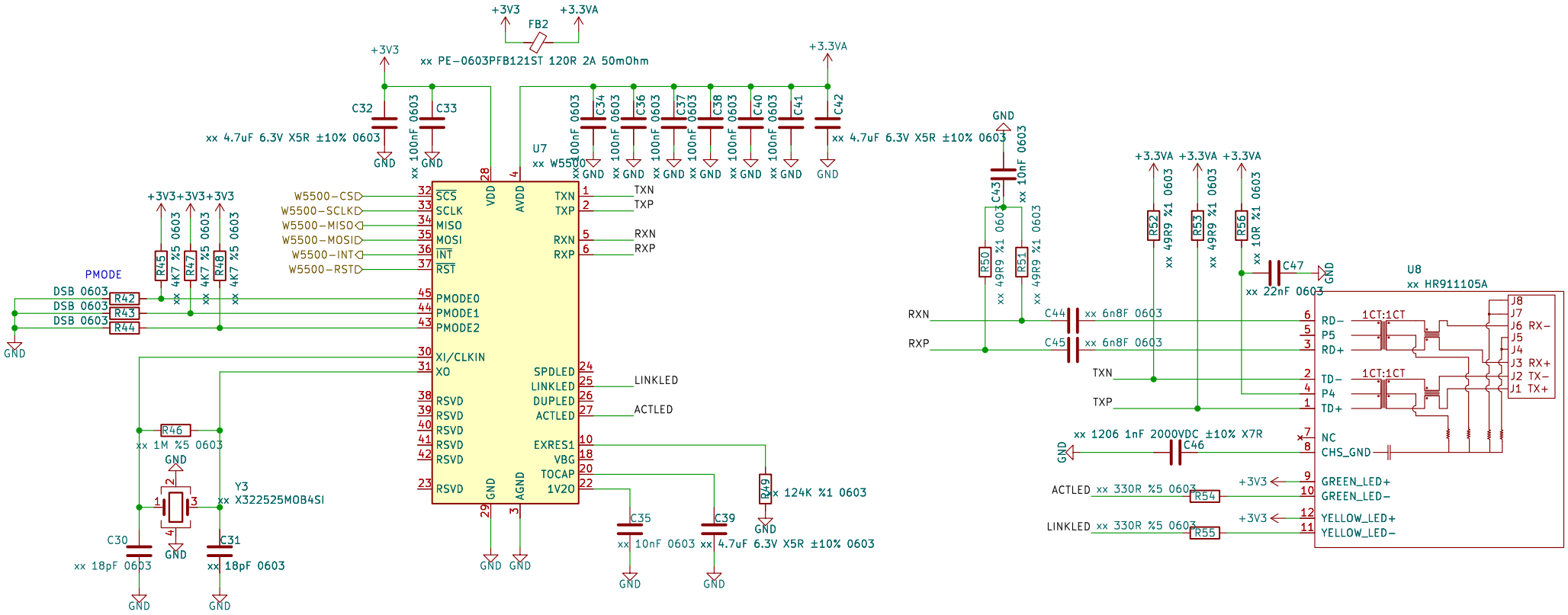
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Size: A4	Date:	**Rev:**
KiCad E.D.A. kicad 7.0.7-7.0.7~ubuntu20.04.1		Id: 1/8


VDD-EXT
C3
100nF 0603
GND
+3V3
C8
100nF 0603
GND
U2
TXB0104PW
OE
VCCA
VCCB
GSM-RXD-S
GSM-TXD-S
GSM-STATUS
GSM-RXD
GSM-TXD
GSM-STS
M1
A7682E
L-KLS1-SMA002-B
J3
RF_ANT
GND
GSM-TXD-S
GSM-RXD-S
UART1_TXD
UART1_RXD
UART1_RTS
UART1_CTS
UART1_DCD
UART1_DTR
UART1_RI
UART3_TXD
UART3_RXD
GSM-USB-VBUS
GSM-USB-DP
GSM-USB-DM
USB_VBUS
USB_DP
USB_DM
SDA
SCL
DEBUG_RXD
DEBUG_TXD
DBG_RXD
DBG_TXD
VBAT
ADC
PWRKEY
VDD_EXT
NETLIGHT
STATUS
GSM-BOOT-CFG
A7682-RESET
BOOT_CFG
RESET
SIM1_DET
SIM1_DATA
SIM1_CLK
SIM1_RST
SIM1_VDD
SIM-1-DET
SIM-1-DATA
SIM-1-CLK
SIM-1-RST
SIM-1-VDD
SIM2_DET
SIM2_DATA
SIM2_CLK
SIM2_RST
SIM2_VDD
SIM-2-DET
SIM-2-DATA
SIM-2-CLK
SIM-2-RST
SIM-2-VDD
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
MICP
MICN
SPK1P
SPK1N
+BATT
FB1
BLM21PG300SN1D
D1
xx PESD4V0Y1BSF
C13
C14
C15
C16
C17
100uF 6.3V CASE-B 3528
4.7uF 6.3V X5R ±10% 0603
100nF 0603
33pF 0603
U1
xx SkySIM
SIM-2-DATA
DSB 0603
R3
R4
C2
R7
47K %5 0603
SIM-2-DET
R8
C9
C10
C11
C12
xx 33pF 0603
xx 100nF 0603
R9
R5
R6
0R 0603
SIM-1-DETECT
SIM-1-DET
SIM-1-DATA
SIM-1-CLK
SIM-1-RST
SIM-1-VDD
J1
LEY01003-S002A-61S
SW2
SW1
C7
C3
C2
C1
CS
SIM-DET
SIM-DATA
SIM-CLK
SIM-RST
SIM-VDD
CASE
C1
C4
C5
C6
C7
U3
TPSMS05T1G
J2
xx Conn_02x05_Odd_Even
R15
XX 0R 0603
R14
D2
LED_603 GREEN
R12
R13
Q3
BC846
PWRKEY
GSM-PWRKEY
R1
R2
Q1
BC846
A7682-RESET
GSM-RESET
R10
R11
Q2
BC846
NETLIGHT
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U4
STM32G0B0RET6
PC10 PB9 PB8 PB7 PB6 PB5 PB4 PB3 PD6 PD5 PD4 PD3 PD2 PD1 PD0 PC9
RS485-2-TXD LED-1 LED-2 LED-3 FLASH_2_CS FLASH_SPI_MOSI FLASH_SPI_MISO FLASH_SPI-SCK LED-4 SIM-1-LED-1 SIM-1-LED-2 LED-5 SIM-1-LED-3 SIM-1-LED-4 SIM-1-LED-5 ROLE-2-OUT
1 PC11 RS485-2-RXD
2 PC12 DIGITAL-IN-3
3 PC13 FLASH_1_CS
4 PC14-OSC32-IN OSC32-IN
5 PC15-OSC32-OUT OSC32-OUT
6 VBAT
7 VREF+ +3V3
8 VDD/VDDA +3V3
9 VSS/VSSA
10 PF0-OSC-IN OSC-IN
11 PF1-OSC-OUT OSC-OUT
12 NRST NRESET
13 PC0 SIM-1-DETECT
14 PC1 BUTTON
15 PC2 W5500-MISO
16 PC3 W5500-MOSI
48 PC8 ROLE-1-OUT
47 PA15 RS485-1-DIR
46 PA14-BOOT0 SWCLK
45 SWDIO
44 PA12[PA10] POWER-DETECT
43 PA11[PA9] DIGITAL-IN-5
42 PA10 EXP-1-SDA
41 PD9 DIGITAL-IN-1
40 PD8 STS-LED-1
39 PC7 STS-LED-2
38 PC6 EXP-1-RESET
37 PA9 EXP-1-SCL
36 PA8 DIGITAL-IN-2
35 PB15 DIGITAL-IN-6
34 PB14 GSM-PWRKEY
33 PB13 DIGITAL-OUT-2
17 PA0 ANALOG-IN-1
18 PA1 ANALOG-IN-2
19 PA2 GSM-RXD
20 PA3 GSM-TXD
21 PA4 GSM-STS
22 PA5 GSM-RESET
23 PA6 W5500-RST
24 PA7 W5500-INT
25 PC4 CFG-TX
26 PC5 CFG-RX
27 PB0 RS485-1-RXD
28 PB1 RS485-2-DIR
29 PB2 RS485-1-TXD
30 PB10 W5500-SCLK
31 PB11 W5500-CS
32 PB12 DIGITAL-OUT-1
Y2
32768KHz SMD3215-2P
C20 18pF 0603
C22 18pF 0603
BT1 MYOUNG MY-2032-08
C21 4.7uF 6.3V X5R ±10% 0603
C24 4.7uF 6.3V X5R ±10% 0603
C25 100nF 0603
C26 100nF 0603
Y1
X322525MOB4SI
C19 18pF 0603
C23 18pF 0603
R17 100K %5 0603
SW1
TL9210AF180Q
R21 4K7 %5 0603
C28 100nF 0603
BUTTON
MODEL OF BASE
DIGITAL INPUT 6
DIGITAL OUTPUT 2
ANALOG INPUT 2
RELAY OUTPUT 2
USE EXPANDER IC U5
DIGITAL INPUT 8
RELAY OUTPUT 4
GENERAL PURPOSE IO 4
U5
PCA9557PW
R18 2K2 %5 0603
R19 2K2 %5 0603
R20 2K2 %5 0603
C27 100nF 0603
EXP-1-SCL
EXP-1-SDA
EXP-1-RESET
SCL SDA RESET VDD VSS
A0 A1 A2
IO0 IO1 IO2 IO3 IO4 IO5 IO6 IO7
EX-IO-2
ROLE-4-OUT
DIGITAL-IN-8
DIGITAL-IN-4
DIGITAL-IN-7
ROLE-3-OUT
EX-IO-1
R16 4K7 %5 0603
C18 100nF 0603
NRESET
J4
xx 1,27mm 02x05 pin header
PROGRAMMER
SWCLK
SWDIO
NRESET
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Title:
Size: A4
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560R %5 0603
R26
D3
LED_603 GREEN
LED-1
560R %5 0603
R22
D4
LED_603 GREEN
LED-2
560R %5 0603
R23
D5
LED_603 GREEN
LED-3
560R %5 0603
R24
D6
LED_603 GREEN
LED-4
560R %5 0603
R25
D7
LED_603 GREEN
LED-5
GND
+3V3
xx 2K2 %5 0603
R27
xx 2K2 %5 0603
R28
xx 2K2 %5 0603
R29
C29
xx 100nF 0603
GND
U6
xx PCA9557PW
EXP-1-SCL
EXP-1-SDA
EXP-1-RESET
SCL
SDA
RESET
A0
A1
A2
VDD
VSS
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
SIM-2-LED-3
SIM-2-LED-4
SIM-2-LED-1
SIM-2-LED-2
SIM-2-LED-5
GND
GND
SIM-1-LED-1
SIM-2-LED-1
560R %5 0603
R36
xx 560R %5 0603
R37
LED_603 GREEN
D10
D11
xx LED_603 BLUE
SIM-1-LED-2
SIM-2-LED-2
560R %5 0603
R38
xx 560R %5 0603
R39
LED_603 GREEN
D12
D13
xx LED_603 BLUE
SIM-1-LED-3
SIM-2-LED-3
+3V3
R95
xx 1K %5 0603
560R %5 0603
R32
xx 560R %5 0603
R33
LED_603 GREEN
D14
D15
xx LED_603 BLUE
SIM-1-LED-4
SIM-2-LED-4
560R %5 0603
R40
xx 560R %5 0603
R41
LED_603 GREEN
D16
D17
xx LED_603 BLUE
SIM-1-LED-5
SIM-2-LED-5
560R %5 0603
R34
xx 560R %5 0603
R35
LED_603 GREEN
D18
D19
xx LED_603 BLUE
GND
D8
LED_603 GREEN
560R %5 0603
R31
STS-LED-1
GND
560R %5 0603
R30
STS-LED-2
GND
LED_603 RED
D9
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File: LEDS.kicad_sch
Title:
Size: A4
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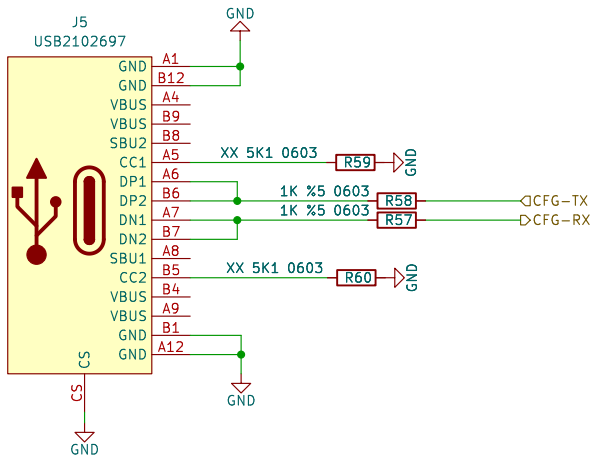


PMODE[2:0]			DESCRIPTION
2	1	0	
0	0	0	10BT Half dublex, Auto negatiation disabled
0	0	1	10BT Full dublex, Auto negatiation disabled
0	1	0	100BT Half dublex, Auto negatiation disabled
0	1	1	100BT Full dublex, Auto negatiation disabled
1	0	0	100BT Half dublex, Auto negatiation enabled
1	0	1	Not used
1	1	0	Not used
1	1	1	All capable, Auto negatiation enabled

Sheet: /Eth/
File: Eth.kicad_sch

Title:

Size: A4	Date:	**Rev:**
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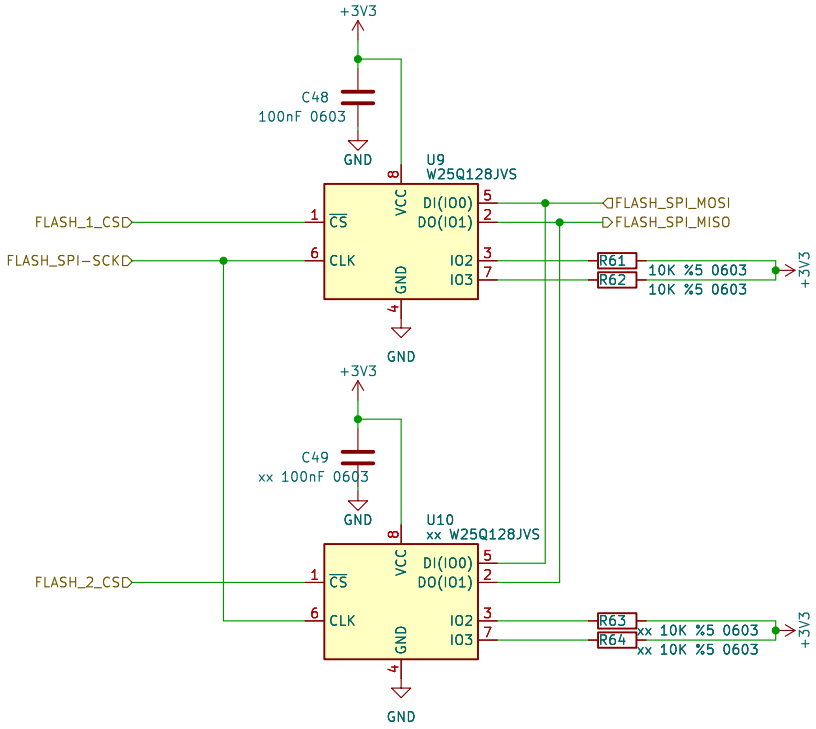


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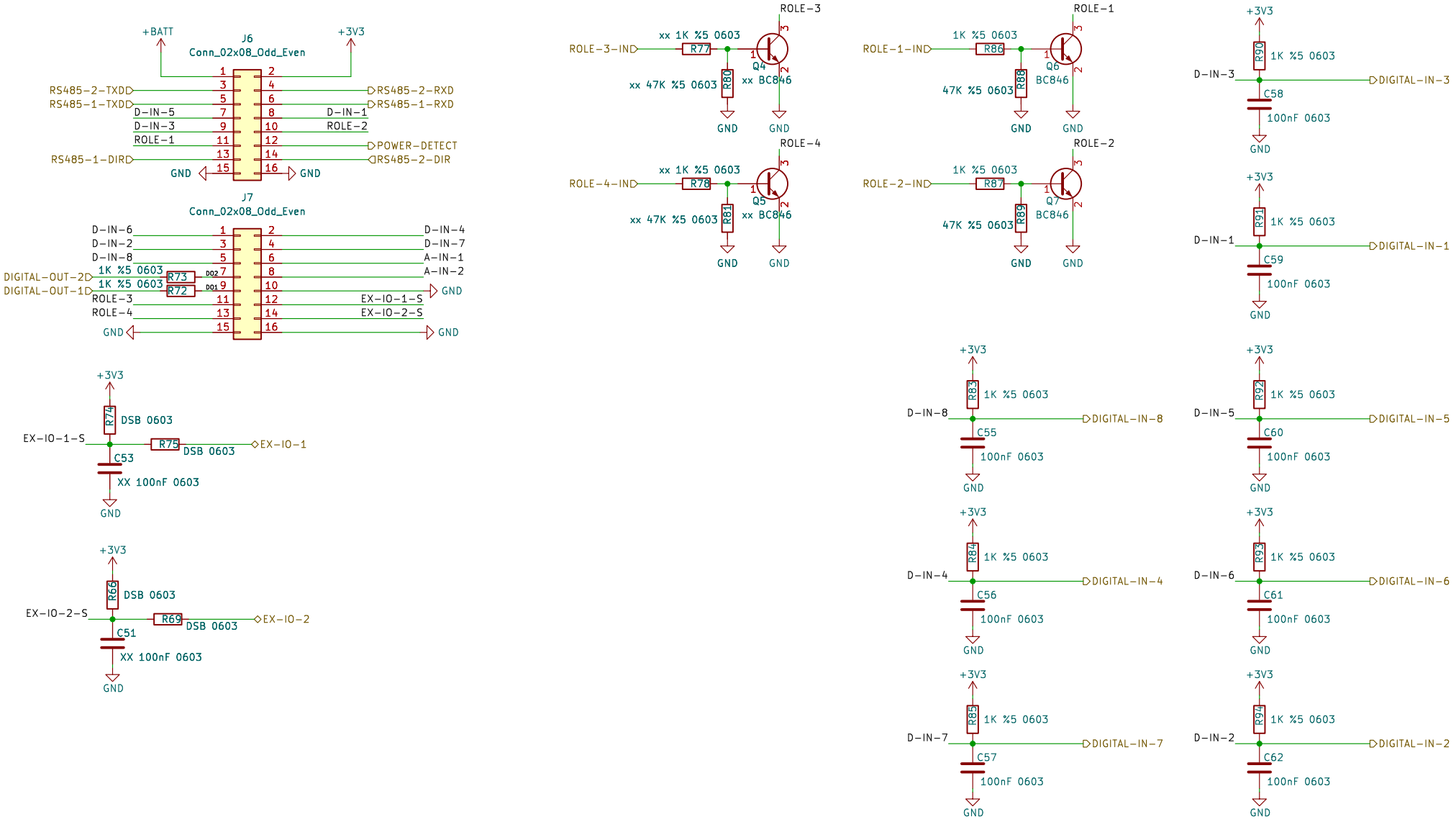
W25Q128 128MBit 16MByte 0,5$
W25Q256 256MBit 32MByte 1,43$



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Size: A4	Date:	**Rev:**
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+BATT
+3V3
J6
Conn_02x08_Odd_Even
RS485-2-TXD
RS485-1-TXD
D-IN-5
D-IN-3
ROLE-1
RS485-1-DIR
GND
RS485-2-RXD
RS485-1-RXD
D-IN-1
ROLE-2
POWER-DETECT
RS485-2-DIR
GND
J7
Conn_02x08_Odd_Even
D-IN-6
D-IN-2
D-IN-8
1K %5 0603
1K %5 0603
DIGITAL-OUT-2
DIGITAL-OUT-1
ROLE-3
ROLE-4
GND
R73
R72
D-IN-4
D-IN-7
A-IN-1
A-IN-2
GND
EX-IO-1-S
EX-IO-2-S
GND
ROLE-3
xx 1K %5 0603
ROLE-3-IN
R77
Q4
xx 47K %5 0603
R80
xx BC846
GND
GND
ROLE-4
xx 1K %5 0603
ROLE-4-IN
R78
Q5
xx 47K %5 0603
R81
xx BC846
GND
GND
ROLE-1
1K %5 0603
ROLE-1-IN
R86
Q6
47K %5 0603
R88
BC846
GND
GND
ROLE-2
1K %5 0603
ROLE-2-IN
R87
Q7
47K %5 0603
R89
BC846
GND
GND
+3V3
R90
1K %5 0603
D-IN-3
DIGITAL-IN-3
C58
100nF 0603
GND
+3V3
R91
1K %5 0603
D-IN-1
DIGITAL-IN-1
C59
100nF 0603
GND
+3V3
R74
DSB 0603
EX-IO-1-S
R75
DSB 0603
EX-IO-1
C53
XX 100nF 0603
GND
+3V3
R66
DSB 0603
EX-IO-2-S
R69
DSB 0603
EX-IO-2
C51
XX 100nF 0603
GND
+3V3
R83
1K %5 0603
D-IN-8
DIGITAL-IN-8
C55
100nF 0603
GND
+3V3
R84
1K %5 0603
D-IN-4
DIGITAL-IN-4
C56
100nF 0603
GND
+3V3
R85
1K %5 0603
D-IN-7
DIGITAL-IN-7
C57
100nF 0603
GND
+3V3
R92
1K %5 0603
D-IN-5
DIGITAL-IN-5
C60
100nF 0603
GND
+3V3
R93
1K %5 0603
D-IN-6
DIGITAL-IN-6
C61
100nF 0603
GND
+3V3
R94
1K %5 0603
D-IN-2
DIGITAL-IN-2
C62
100nF 0603
GND
1K %5 0603
383R %1 0603
ANALOG-IN-1
R67
R71
A-IN-1
C52
100nF 0603
R70
124R %1 0603
GND
GND
1K %5 0603
383R %1 0603
ANALOG-IN-2
R76
R82
A-IN-2
C54
100nF 0603
R79
124R %1 0603
GND
GND
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